Free variables are unchanged by substituting for a variable that does not occur freely

The Open Logic Project contributors

openlogic:lambda-calculus/syntax:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-syntax/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the abstraction chain is verbatim with each of its four steps annotated by the author with the cited clause or hypothesis; the variable and application cases carry the author's exercise markers for the two immediate unfoldings.

Editorial note. The label thm:notinfv names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Substitution clause for abstractions. Same unit, \ollabel{defn:substitution-4} in the substitution definition. First line of the abstraction chain (line 1250).
- Inductive definition of free variables. Syntax unit, Definition of FV, clause fv2 for abstraction. Lines 1251-1254: unfold FV of an abstraction as FV of the body minus the bound variable, used in both directions of the chain.

Source reference. Original definition defn-substitution-4: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1203–1203. Prerequisite pointer invoked at proof line(s) 1253; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition def-fv: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 1255; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition def-fv2: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1116–1116. Prerequisite pointer invoked at proof line(s) 1255; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *If $x \notin \mathrm{FV}\, M$, then $\mathrm{FV}\, M[N/x] = \mathrm{FV}\, M$, if the left-hand side is defined.*

Proof. By induction on the formation of M.

1. M is a variable: exercise.
2. M is of the form (PQ): exercise.
3. M is of the form $\lambda[y][P]$, and since $\lambda[y][P][N/x]$ is defined, it has to be $\lambda[y][P[N/x]]$. Then $P[N/x]$ has to be defined; also, $x \neq y$ and $x \notin \mathrm{FV}\, Q$. Then:

$$
\begin{aligned}
\mathrm{FV}\, \lambda[y][P][N/x] = & \\
& = \mathrm{FV}\, \lambda[y][P[N/x]] && \text{by defn-substitution-4} \\
& = \mathrm{FV}\, P[N/x] \setminus \{y\} && \text{by def-fvdef-fv2} \\
& = \mathrm{FV}\, P \setminus \{y\} && \text{by inductive hypothesis} \\
& = \mathrm{FV}\, \lambda[y][P] && \text{by def-fvdef-fv2}
\end{aligned}
$$

□